

Maha Marakkalage Sanchila Amavi

 sanchila-amavi |  sanchila-amavi |  Portfolio |  sanchilaamavi@gmail.com |  +94 76 422 7785

Professional Summary

Electronics and Telecommunication Engineering undergraduate at General Sir John Kotelawala Defence University with strong interests in Embedded Systems, Artificial Intelligence, Machine Learning, Computer Vision, IoT, Wireless Communication, Robotics and Full-Stack Software Development. Experienced in designing and developing end-to-end engineering solutions spanning embedded hardware, cloud-connected IoT systems, AI-powered applications, mobile applications, web platforms and autonomous robotic systems. Proficient in Python, C/C++, Dart, Flutter, React.js, Firebase, OpenCV, YOLOv8 and embedded system development. Skilled in PCB design, firmware development, cloud integration, API development, mobile and web application deployment, data analysis and AI model development. Demonstrates strong analytical, problem-solving, teamwork and project management skills through multidisciplinary engineering and research projects.

Education

General Sir John Kotelawala Defence University <i>BSc. (Hons) in Electronic and Telecommunication Engineering</i>	Feb. 2024 - Present <i>CGPA: 3.32/4.00</i>
Devi Balika Vidyalaya, Colombo 08 <i>GCE Advanced Level (Physical Science Stream)</i> <i>GCE Ordinary Level</i>	Jan. 2014 - Dec. 2022 <i>3 Cs</i> <i>9 As</i>

Relevant Coursework

- Electronic Systems - Signals and Systems - Communication Theory - Electromagnetics	- Control Systems - Electronic Circuits - Communication Networks - Image Processing and Vision	- Digital Signal Processing - Embedded Systems - Power Electronics - Digital System Design
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Technical Skills

Programming	Python, C, C++, Dart, JavaScript, HTML5, CSS3, SQL
AI/ML & Vision	Machine Learning, Deep Learning, Computer Vision, OpenCV, YOLOv8, PyTorch, TensorFlow, Keras, CNNs, MobileNetV2, Transfer Learning, Object Detection, Image Processing, MediaPipe, ONNX Runtime, PERCLOS Analysis
Data & Analysis	NumPy, Pandas, Matplotlib, Data Visualization, Statistical Analysis
Embedded Systems, IoT & Robotics	ESP32, ESP32-S3, Arduino, Embedded C/C++, Sensor Integration, LoRa (SX1278), GSM (SIM800L), Motor Control, PCB Design, IoT Systems Development, Autonomous Navigation, Path Planning, Sensor Fusion, Obstacle Detection, PID Control
Mobile, Web & Cloud	Flutter, Dart, React.js, Tailwind CSS, FastAPI, REST APIs, Firebase Realtime Database, FCM, Firebase Hosting, Firebase Storage, Docker, Android Development
Communication	LoRa, GSM, Wi-Fi, UART, SPI, I ² C
Design Tools	EasyEDA, Proteus, LTspice, SolidWorks, MATLAB, Arduino IDE, VS Code, Microchip Studio, Git, GitHub, Docker, Postman, Canva

Projects

Mine Pulse Smart Underground Mine Safety Monitoring and Warning System	GitHub
- Designed and developed a multi-node IoT safety system for real-time underground hazard detection covering methane (MQ-4), carbon monoxide (MQ-7) and flooding across multiple mine sites. - Designed custom PCBs for both underground and surface nodes. - Implemented a multi-node LoRa protocol with collision avoidance, unique node addressing and ACK handshaking; a single surface gateway supports multiple underground nodes simultaneously. - Built Firebase Realtime Database cloud integration with hazard uploads, alert history and real-time push notifications via FCM to a Flutter mobile app and live web dashboard. - Delivered SMS alerts via GSM on hazard detection while simultaneously triggering OLED hazard display, audible buzzer alarm and cloud notification pipeline.	
Technologies: ESP32-S3, LoRa SX1278, SIM800L, MQ-4/MQ-7, Firebase, Flutter, React.js, EasyEDA, C++, Dart	

Driver Fatigue Detection System | AI-Powered Driver Safety Application

[GitHub](#)

- Designed and developed a full-stack real-time driver safety platform combining deep learning fatigue detection, live GPS tracking, voice alerts and emergency response features.
- Trained and optimized an EfficientNet-B0 CNN model for four-class behavioural classification (open eye, closed eye, yawn, no-yawn) with CLAHE preprocessing and Albumentations augmentation.
- Implemented PERCLOS-based continuous fatigue scoring with an intelligent alert engine that triggers escalating voice alerts when drowsiness thresholds are exceeded.
- Built a Flutter mobile application featuring real-time camera-based face detection, a live GPS map widget with Google Maps integration, an emergency contact management system and push notification support.
- Developed a FastAPI backend with JWT authentication, trip history recording, AI inference endpoints and SQLAlchemy database management; containerised with Docker for production deployment.
- Integrated Firebase services for authentication, cloud messaging and real-time data synchronisation across mobile and backend layers.

Technologies: Python, PyTorch, OpenCV, FastAPI, Flutter, Dart, Google Maps, Firebase, Docker, REST APIs

LankaMesh | LoRa-Based Decentralized Mesh Communication Network

[GitHub](#)

- Designed and developed a decentralized wireless mesh communication network using LoRa technology for emergency and low-connectivity environments.
- Implemented LoRa-based long-range peer-to-peer communication for SOS alerts, messages and GPS data sharing.
- Designed embedded system architecture using ESP32-S3, LoRa RA-02 (SX1278) and GPS modules.
- Built a Flutter mobile application for emergency alert categorization and communication handling.
- Demonstrated scalable IoT-based disaster communication framework for resilient connectivity.

Technologies: ESP32-S3, LoRa RA-02, GPS, Flutter, IoT, Embedded Systems, Wireless Communication

AI Proctoring System | Intelligent Real-Time Examination Monitoring Platform

[GitHub](#)

- Developed a real-time exam proctoring platform combining MediaPipe face detection and YOLOv8 object detection to automatically identify violations including absent candidates, multiple persons and unauthorised device use.
- Implemented a dynamic risk scoring engine with configurable thresholds, audio alerts and a live dashboard featuring risk-over-time graphs and event distribution analytics.
- Architected a multi-platform system with a Python AI inference core, FastAPI web backend and Flutter mobile application to support cross-platform remote supervision.
- Delivered comprehensive session logging to CSV with timestamps, event metadata and risk summaries for post-examination audit and review.

Technologies: Python, YOLOv8, MediaPipe, OpenCV, FastAPI, Flutter, Dart, Tkinter, Matplotlib, C++

Mars Robot Competition | Autonomous Robotics System

[GitHub](#)

- Designed and developed an autonomous robotic system for competition-based navigation and object handling tasks.
- Implemented sensor fusion using IR sensors, ultrasonic sensors, gyroscope and color sensors.
- Developed embedded control system using ESP32-S3 for real-time decision-making and motor control.
- Designed mechanical system using SolidWorks and 3D printing for structural fabrication.
- Implemented encoder-based motor control using TB6612FNG driver for precise movement control.

Technologies: ESP32-S3, C/C++, IR, Ultrasonic, MPU6050, TCS34725, SolidWorks, Embedded Systems

ROSCO'25 Competition Robot | Autonomous Line and Wall Following Robot

[GitHub](#)

- Designed and developed an autonomous robot for the ROSCO'25 competition supporting black/white line following, wall following, obstacle avoidance and ramp navigation tasks.
- Implemented sensor fusion across an 8-channel IR array, VL53L0X ToF sensors and MPU6050 IMU with a modular state-machine architecture separating perception, decision and control layers.
- Developed PID-based control algorithms on ESP32-S3 with TB6612FNG motor driver for line following accuracy, wall distance regulation and ramp pitch stabilisation.
- Designed chassis using SolidWorks, fabricated via laser-cut acrylic and 3D-printed custom sensor and motor mounts.

Technologies: ESP32-S3, VL53L0X, IR Sensors, MPU6050, TB6612FNG, SolidWorks, Embedded Systems

- Developing a portable assistive device for blind users to measure liquid levels using non-contact capacitive sensing.
- Implementing external capacitive detection through cup walls for real-time liquid level monitoring without liquid contact.
- Using ESP32-C3 Mini for sensor processing, calibration and multi-level detection.
- Designing vibration and buzzer feedback system with different alerts for fill levels.
- Building a compact clip-on device with rechargeable battery and low-power embedded design.

Technologies: ESP32-C3 Mini, Capacitive Sensing, Embedded Systems, PCB Design, IoT, Vibration Motor, Buzzer

Achievements

Winner(Inter University/Female)	MSSC Shooting Championship - 2025
Champions	Target Sprint Shooting Competition - 2025
KDU Colours	Air Rifle - 2025
Precision Shooter Award	MSSC Shooting Championship - 2024
Sharp Shooter Award	Inter University Shooting Championship - 2024
KDU Merit	Air Rifle - 2024

Certifications

AI/ML Engineer	2026, SLIIT; Machine learning, supervised and unsupervised learning, deep learning, neural networks, TensorFlow, Keras and model evaluation techniques.
PCB Designing Workshop	2025, IET KDU & Sri Lanka Institute of Robotics (SLIR); Schematic design, component placement, PCB layout design, design rule checking (DRC), Gerber file generation and PCB manufacturing workflow.
MATLAB Fundamentals	2024, MATLAB Training Program; Numerical computing, matrix operations, data visualization, scripting and engineering simulation techniques.

Leadership & Volunteering

Secretary: Shooting Sports Association KDU	August 2025 – Present
Finance Team Member: GENESIZ'26	May 2026 – Present
Finance Committee Member: ERIC KDU	Feb. 2026 – Present
Program Team Member: IEEE ComSoc KDU	June 2024 – Present
Design Committee Member: Mathematical Society KDU	Feb. 2025 – Feb. 2026
PR Team Member: ERIC KDU	Feb. 2025 – Feb. 2026

References

Prof. S.U. Dampage Professor Dept. of Electronic and Telecom. Engineering General Sir John Kotelawala Defence University Email: dampage@kdu.ac.lk Tele: +94 70 340 5405	ENG. (Mrs.) P.N. Karunanayake Senior Lecturer Dept. of Electronic and Telecom. Engineering General Sir John Kotelawala Defence University Email: karunanayake@kdu.ac.lk Tele: +94 77 333 1189
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